

RANGEDALA, Sweden

WMO: 025360

Lat: 57.7846N

Lon: 13.1642E

Elev: 299

StdP: 97.78

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-11.0	-15.4	1.0	-11.5	-13.2	1.2	-9.0	8.7	0.2	8.0	0.7	1.1	20	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	25.1	16.9	23.5	16.3	21.7	15.6	18.5	22.5	17.4	21.5	16.5	20.2	2.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.6	19.9	16.0	11.7	18.8	15.0	11.0	17.6	53.4	22.3	50.1	21.7	47.2	20.2	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.2	5.4	DB	-16.1	27.7	2.8	1.6	-18.1	28.9	-19.7	29.8	-21.3	30.7	-23.4	31.9	(4)
(5)			WB	-16.4	20.4	2.7	1.4	-18.3	21.5	-19.9	22.3	-21.4	23.1	-23.4	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-2.5	-2.0	0.6	5.5	10.0	13.3	15.7	15.0	11.2	6.4	2.2	-0.7	(6)	
	DBStd	7.38	4.32	4.15	3.89	3.79	3.77	3.09	2.89	2.73	2.75	3.46	3.70	4.41	(7)	
	HDD10.0	1911	389	336	291	143	47	4	0	1	19	117	233	331	(8)	
	HDD18.3	4420	647	569	549	384	259	156	92	108	214	369	483	589	(9)	
	CDD10.0	553	0	0	0	8	47	102	178	157	55	6	0	0	(10)	
	CDD18.3	21	0	0	0	0	1	3	11	6	0	0	0	0	(11)	
	CDH23.3	155	0	0	0	1	7	34	87	26	0	0	0	0	(12)	
	CDH26.7	13	0	0	0	0	0	2	10	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.6	2.8	2.9	2.9	2.8	2.5	2.2	2.0	2.1	2.3	2.7	3.0	2.9	(14)	
(15) Precipitation	PrecAvg	942	86	60	52	56	60	85	87	90	86	92	87	91	(15)	
	PrecMax	1155	182	173	101	123	116	149	180	175	203	206	155	194	(16)	
	PrecMin	651	11	16	14	6	3	0	13	33	20	0	34	15	(17)	
	PrecStd	140	46	38	23	29	25	35	40	38	51	45	31	48	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	6.9	13.0	19.6	24.3	26.5	28.0	25.5	21.2	15.2	10.6	8.3	(19)	
		MCWB	5.9	5.5	7.0	12.0	15.0	16.7	18.2	17.7	15.8	12.5	10.0	8.0	(20)	
	2%	DB	5.0	5.3	9.9	16.8	21.7	24.1	25.7	23.9	18.9	13.0	9.2	6.9	(21)	
		MCWB	4.7	4.6	5.9	10.0	13.9	15.5	17.4	17.3	14.7	12.1	8.7	6.7	(22)	
	5%	DB	3.9	4.4	7.3	14.3	19.4	21.9	23.8	22.0	16.9	11.9	8.2	6.0	(23)	
		MCWB	3.6	3.7	4.8	8.9	12.5	14.7	16.8	16.5	13.6	10.9	7.8	5.6	(24)	
	10%	DB	2.9	3.3	5.4	11.7	16.9	19.4	21.8	20.0	15.3	11.0	7.2	4.9	(25)	
		MCWB	2.6	2.7	3.6	7.4	11.4	13.7	16.2	15.9	13.0	10.1	6.8	4.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	6.2	8.0	13.1	16.6	18.4	19.9	19.5	16.6	13.4	10.2	8.1	(27)	
		MCDB	6.1	6.6	11.6	18.6	22.2	23.7	24.4	22.8	19.5	14.2	10.5	8.2	(28)	
	2%	WB	4.8	5.0	6.5	10.9	14.8	16.6	18.7	18.2	15.2	12.2	8.9	6.7	(29)	
		MCDB	4.9	5.2	9.1	15.3	19.5	21.9	23.1	21.5	17.8	12.8	9.1	6.9	(30)	
	5%	WB	3.7	3.8	5.1	9.3	13.5	15.4	17.8	17.2	14.2	11.2	7.8	5.7	(31)	
		MCDB	3.8	4.2	6.8	13.0	17.7	20.0	22.1	20.5	16.1	11.8	8.1	5.9	(32)	
	10%	WB	2.7	2.8	4.0	8.0	12.1	14.3	16.7	16.3	13.3	10.3	6.8	4.6	(33)	
		MCDB	2.9	3.2	5.1	11.1	16.2	18.5	20.8	19.3	15.0	10.8	7.2	4.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	4.1	6.0	8.0	8.7	8.4	8.1	7.3	6.5	4.9	3.5	3.6	(35)	
		MCDBR	3.9	4.5	8.9	11.6	12.1	12.1	12.0	10.3	8.5	5.2	4.1	3.9	(36)	
	5% WB	MCWBR	4.0	3.8	5.7	6.4	6.4	5.8	5.4	5.0	4.9	4.1	4.0	4.1	(37)	
		MCWBR	3.9	3.9	7.7	10.1	10.8	10.5	9.9	8.5	7.2	4.9	3.8	3.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.281	0.312	0.329	0.367	0.364	0.354	0.375	0.372	0.342	0.331	0.302	0.266	(40)	
		taud	2.276	2.347	2.379	2.335	2.400	2.447	2.402	2.414	2.489	2.442	2.334	2.210	(41)	
	Edn at Noon		583	732	824	844	871	885	854	831	806	703	550	474	(42)	
		Edh at Noon	47	69	87	106	107	103	106	98	78	63	45	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.37	0.97	2.41	3.88	5.10	5.58	5.07	4.00	2.77	1.27	0.47	0.25	(44)	
	RadStd		0.07	0.16	0.24	0.52	0.47	0.36	0.66	0.39	0.34	0.21	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page